

Comprehensive Guide to LaTeX Tables

Reference Guide

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1 Introduction to Tables in LaTeX

This document demonstrates various table creation techniques in $\text{\LaTeX}$, ranging from simple tabular environments to advanced features like multi-page tables, color styling, and cell merging.

2 Basic Table

A simple table without borders, using basic column alignment.

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

3 Table with Borders

Adding vertical borders improves readability and structure.

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

4 Advanced Table with Double Borders and Horizontal Lines

A more complex table featuring double vertical borders, horizontal lines, and spacing control.

Col1	Col2	Col3	Col4
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

5 Table with Caption and Label

Tables can include captions for reference and labels for cross-referencing within the document.

Table 1 demonstrates a basic table with a caption.

Table 1: Data Summary Table

Col1	Col2	Col3	Col4
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

6 Table with Column Merging

Using `\multicolumn` to merge columns for a unified header.

Table 2: Country Codes Reference

Country List			
Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO Numeric Code
Afghanistan	AF	AFG	004
Åland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

7 Multi-Page Table with Longtable

The `longtable` environment allows tables to span multiple pages, ideal for large datasets.

Table 3: Long Table Example

Beginning of Table	
Item	Value
Row 1	Data 1
Row 2	Data 2
Row 3	Data 3
Row 4	Data 4
Row 5	Data 5
Row 6	Data 6
Row 7	Data 7
Row 8	Data 8
Row 9	Data 9
Row 10	Data 10
Row 11	Data 11
Row 12	Data 12
Row 13	Data 13
Row 14	Data 14
Row 15	Data 15
End of Table	

8 Alternating Row Colors

Using `\rowcolors` to create visually appealing striped tables.

Table 4: Countries with Alternating Row Colors

Country List		
Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code
Afghanistan	AF	AFG
Åland Islands	AX	ALA
Albania	AL	ALB
Algeria	DZ	DZA
American Samoa	AS	ASM
Andorra	AD	AND
Angola	AO	AGO

9 Advanced Color Styling

Combining cell, row, and column coloring with custom color definitions.

Table 5: Colorful Table Example

Country List		
Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code
Afghanistan	AF	AFG
Åland Islands	AX	ALA
Albania	AL	ALB
Algeria	DZ	DZA
American Samoa	AS	ASM
Andorra	AD	AND
Angola	AO	AGO

10 Professional Table with booktabs

Using the `booktabs` package for publication-quality tables.

Table 6: Professional Booktabs Style

Country	Population (M)	Area (km ²)	Density
Kenya	53.77	580,367	92.6
Tanzania	61.50	947,300	64.9
Uganda	45.74	241,038	189.8
Rwanda	13.25	26,338	503.1
Ethiopia	117.88	1,104,300	106.7

Summary of Useful Commands

Table 7: Common Table Commands Reference

Command	Description
<code>&</code>	Column separator
<code>\\</code>	New row
<code>\hline</code>	Horizontal line across all columns
<code>\cline{i-j}</code>	Partial horizontal line from column i to j
<code>\multicolumn{n}{fmt}{text}</code>	Merge n columns with custom formatting
<code>\multirow{n}{width}{text}</code>	Merge n rows (requires multirow package)
<code>\rowcolor{color}</code>	Color entire row
<code>\cellcolor{color}</code>	Color individual cell
<code>\columncolor{color}</code>	Color entire column
<code>\toprule</code>	Top rule (booktabs)
<code>\midrule</code>	Middle rule (booktabs)
<code>\bottomrule</code>	Bottom rule (booktabs)